

Book Of Proof.

3.10) 2) $1 + 2 + 3 + \dots + n = \binom{n+1}{2}$

let $S = \{0, 1, 2, 3, \dots, n-2, n-1, n\}$ so $|S| = n+1$. The R.H.S counts

2 element subsets of S . The L.H.S can be thought of as a working for loop in programming languages. The 2-element subsets

of S are of form $\{a, b\}$ for $a, b \in S$. When we choose $a = 0$, then

there are n ways to choose b from $1, 2, \dots, n$. Similarly

choose $a = 0$ n ways to choose b

" $a = 1$ $n-1$ " " " ($\because b \neq 0$ as $\{0, 1\}$ already done)

$\vdots$

$a = n-1$ $n - (n-1) = 1$ " "

$a = n$ 0

so L.H.S counts $0 + 1 + \dots + n$ ways to make $\{a, b\}$ from S .

3.10) 4) $P(n, k) = P(n-1, k) + k \cdot P(n-1, k-1)$

Analyze with a small example like $n=3$ & $k=2$ for set $\{1, 2, 3\}$ for

understanding. let $B = \{1, 2, 3, \dots, n\}$ so $|B| = n$. The L.H.S

counts k -permutations of B . The right side divides B into 2

parts. say $\{1\}$ & $\{2, 3, \dots, n\}$. let $B = \{2, 3, \dots, n\}$. Now

$P(n-1, k)$ makes k permutations from B . That is, it makes k permutations

of B that do not include 1. The $P(n-1, k-1)$ makes $k-1$

permutations of B that do not include 1. Then place 1 in

this permutation. There are k positions to place 1 among $k-1$

elements. By multiplication principle, there are $k \cdot P(n-1, k-1)$

k -permutations of B that include 1. By addition principle, there

are total of $P(n-1, k) + k \cdot P(n-1, k-1)$ k -permutations of B .

e.g. if $k-1 = 2$ then there are $k = 3$ ways to place 1 in 2 places e.g. $1 _ 2 _ 3$

3.10) 6) similar to 5. $\binom{3n}{3} = 3\binom{n}{3} + 6n\binom{n}{2} + n^3$.

Supp $|A|=n, |B|=n \text{ \& } |C|=n$. $A \cap B = \emptyset, B \cap C = \emptyset \text{ \& } A \cap C = \emptyset$ so $A \cup B \cup C$
 $= 3n$. L.H.S is 3-subsets of $A \cup B \cup C$. Now for R.H.S, if we

choose all 3 elmts from one of A, B or C , then there are $\binom{n}{3}$ ways,

for all 3^{so} there are $3\binom{n}{3}$ ways if we choose all 3 elmts

from only one set. If we choose 2 from one set e.g. A , then there

are $\binom{n}{2}$ ways to do this & $2n$ ways to select third elmt from B or C .

so total $2n\binom{n}{2}$ ways. similarly for B & C , total $2n\binom{n}{2} + 2n\binom{n}{2} + 2n\binom{n}{2}$

$= 6n\binom{n}{2}$ ways. Finally if we choose one elmt from each set,

there are n^3 combinations. A total of $3\binom{n}{3} + 6n\binom{n}{2} + n^3$ ways.

3.10) 8) $\sum_{k=0}^m \binom{m}{k} \binom{n}{p-k} = \binom{m+n}{m+p}$

L.H.S: $\binom{m}{0}\binom{n}{p-0} + \binom{m}{1}\binom{n}{p-1} + \dots + \binom{m}{m-1}\binom{n}{p-(m-1)} + \binom{m}{m}\binom{n}{p-m}$

select $m-0=m$
 select 0, choose p
 total $m+p$

select $m-1$, choose 1
 select 1, choose $p-1$

select $m-m=0$
 choose m
 select m , choose $p-m$

R.H.S counts $m+p$ sized subsets of $|S|=m+n$. Now divide S into

A & B with $|A|=m$ & $|B|=n$ with $A \cap B = \emptyset$. The above

sketch shows how we can preselect $m-k$ elmts from A & choose

k elmts & ~~choose~~ ^{preselect} k elements from B & choose $p-k$ elements.

This ranges from $k=0$ to m of course.

3.10) 10) This again can be understood better with help of forming

a committee. $\sum_{k=1}^n k\binom{n}{k} = n \cdot 2^{n-1}$

Supp we want to form committ. of any size ≥ 1 & from that committee, we want to choose 1 leader.

R.H.S: From n persons, there are n ways to choose a leader. For rest

$n-1$ persons, each can be in committee or not. This gives 2^{n-1}

ways to choose members. A total of $2^n - 1$ ways.

L.H.S: Suppose we call size = k then $1 \leq k \leq n$. We can choose

k people from n to be in committee. & from those k , select

a leader. First has $\binom{n}{k}$ ways & second has k ways giving $k \binom{n}{k}$

As k can range from 1 to n . It gives $\sum_{k=1}^n k \binom{n}{k}$.

3.10) 12) $\sum_{k=0}^n \binom{n}{k} \binom{k}{m} = \binom{n}{m} 2^{n-m}$.

This is like Q3 except that now committee can be of any size k .

The L.H.S sums this up. The R.H.S choose m leaders first &

then from remaining $n-m$ people, makes a choice to include them in committee or not, that is 2^{n-m} .